File permissions in Linux - Rayane Merlin

Project description

Auditing and updating file and directory permissions within the ~/projects directory

Check file and directory details

```
researcher2@94f6ce76cb82:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Nov  3 18:57 .
drwxr-xr-x 3 researcher2 research_team 4096 Nov  3 19:49 ..
-rw--w---- 1 researcher2 research_team   46 Nov  3 18:57 .project_x.t
xt
drwx--x--- 2 researcher2 research_team 4096 Nov  3 18:57 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Nov  3 18:57 project_k.tx
t
-rw-r----- 1 researcher2 research_team   46 Nov  3 18:57 project_m.tx
t
-rw-rw-r-- 1 researcher2 research_team   46 Nov  3 18:57 project_r.tx
t
-rw-rw-r-- 1 researcher2 research_team   46 Nov  3 18:57 project_t.tx
t
```

Describe the permissions string

Example with project_x.txt :

- First char : **-** (file)
- User : read and write rights
- Group : write rights
- Others : nothing

Change file permissions

The policy states that others are not allowed to have writing rights on files. The file project_k.txt currently does not satisfy this statement. We enter this command to change that :

```
researcher2@94f6ce76cb82:~/projects$ chmod o-w project_k.txt
researcher2@94f6ce76cb82:~/projects$ ls -la project_k.txt
-rw-rw-r-- 1 researcher2 research_team 46 Nov  3 18:57 project_k.txt
```

Change file permissions on a hidden file

We remove writing permissions for everyone on .project_x.txt by using :

```
researcher2@94f6ce76cb82:~/projects$ chmod u-w,g-w .project_x.txt
researcher2@94f6ce76cb82:~/projects$ ls -la .project_x.txt
-r-------- 1 researcher2 research_team 46 Nov  3 18:57 .project_x.txt
```

Change directory permissions

We have to remove all rights for group and other by typing this command :

```
researcher2@94f6ce76cb82:~/projects$ chmod g-x drafts
researcher2@94f6ce76cb82:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Nov  3 18:57 .
drwxr-xr-x 3 researcher2 research_team 4096 Nov  3 19:49 ..
-r-------- 1 researcher2 research_team   46 Nov  3 18:57 .project_x.t
xt
drwx------ 2 researcher2 research_team 4096 Nov  3 18:57 drafts
```

Summary

Three security misconfigurations encountered across standard files like hidden files and subdirectories. These updates brought the research team's file system to a full compliance with given security policies.